

AMANATULLAH PANDU ZENKLINOV

South Jakarta, Indonesia

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INTRODUCTORY PROFILE



As a dedicated educator, researcher, and data scientist, I am an expert in joining theoretical statistics and computer science with practical AI and machine learning applications. My real work is changing complex theoretical terms into easy to understand solutions with special experience in AI model training, benchmarking and deployment. My aim is to use as many tools as possible so that AI can be executed fast taking advantage of my ability to quickly grasp, go for the new technologies and rank the tasks in changing environments. I insist on the importance of providing a clear message to other people and sharing knowledge through the mentoring of students and colleagues who thus will achieve technical excellence. My focus is on having the power of good in precision, interest-driven problem-solving, and ethical AI development, a way of thinking that will enable me to improve the relations between people and machines by going deeper into data and decision-making.

CORE COMPETENCIES AND TECHNOLOGIES

Programming Languages: Python, R, SQL
Frameworks: NLTK, Scikit-Learn, Opencv, Scipy, PyTorch, TensorFlow, Transformer, Langchain, Ollama, Apache Spark, Hadoop
Databases: MongoDB, PostgreSQL, MySQL, Google BigQuery
Cloud Services: Google Cloud Platform (GCP), AWS
Visualization: Tableau, Microsoft Power BI, Google Looker
Statistical Tools: Minitab, IBM SPSS, Matlab, Eviews, GeoDa/ArcGis, Vertex AI
Skills and Area of Expertise: Research, Data Science, Big Data Analysis, Machine Learning (ML), Deep Learning (DL)

WORKING EXPERIENCE

Matana University - Tangerang, Indonesia

● Head of the Statistics Undergraduate Programme - Full-Time (On-Site)

Mar 2025 – Present

Role and Key achievements:

- Create, update, and ensure the curriculum of undergraduate statistics is aligned with academic standards and industry requirements. Directed to Data Science (AI/ML/DL) and Machine Learning in Production.
- Conduct classes, counsel students regarding educational and career goals, and guide thesis or research projects.
- Execute program-related tasks, such as the scheduling of events, the preparing of budgets, the execution of policies, and the maintenance of the program's accreditation.
- Recruit, evaluate, and care for study programme members in order to build up the team and promote the principles of professionalism and cooperation.
- Create partnerships with businesses, involve the alumni in the project, and make the internships and applications for use in the real world known.
- Promote undergraduate research, bring departments together on the topic for a single project, and have study programme members and students take part in projects at the cutting edge of discovery.
- Visualize long-term strategies for the growth of the program, distribution of resources, and making it compatible with other statistics education programs.
- Come up with the best ways to offer assistance to students and create an inclusive environment and take care of the students' welfare and retention through counseling and support systems implementation.

● Quality Assurance Committee of Statistics Department - Full-Time (On-Site)

Jan 2025 – Present

Role and Key achievements:

- Prepare the lecture schedule each semester and develop the curriculum along with supporting documents.
- Conduct monitoring of academic activities, perform evaluations, and implement follow-up actions for continuous academic quality enhancement.
- Submit task reports at the end of each semester and maintain a database for accreditation purposes.

● Lecturer of the Statistics Undergraduate Programme - Full-Time (On-Site)

Aug 2024 – Present

Role and Key achievements:

- Prepared and presented interesting and thorough lecture materials on the topics of data science, statistics, machine learning, or deep learning and used the latest data science methods to produce practical examples and assignments.
- Led students in the process of conducting research projects to improve their AI/ML algorithms and data-driven decision-making skills.
- Courses:
 - Algorithm and Programming in R and Python
 - Database and Data Retrieval in SQL with PostgreSQL and non-RDBMS with MongoDB
 - Database and Data Retrieval in Big Data (HDFS, Spark, Kubernetes, Kafka, Google Query, Heroku, etc)
 - Data Exploration and Data Visualization
 - Introduction to Data Science
 - Multivariate Analysis (CCA, Conjoint Analysis, Path Analysis, Factor Analysis, SEM, MDS, PSM, IV, etc.)
 - Numerical Methods
 - Profession Ethic in Statistics and Data Science
 - Statistical Computation

Role and Key achievements:

- Handled ±100 projects related to data analysis, data science, machine learning, and AI system (Chatbot LLM).
- Applied advanced techniques to clean, transform, and visualize data, e.g., using R & Python libraries such as Pandas and Matplotlib, or applying ETL processes in SQL. Additionally, proficient in creating visualizations using Power BI, Tableau, and Google Looker (Data Studio).
- Developed tailored AI/ML models, including deep learning architectures, e.g., NLP/LLM for sentiment analysis, CNN/RNN for image recognition, dimensional reduction for feature selection, and cluster analysis for market segmentation.
- Optimized model performance, including XGBoost and Decision Trees, through meticulous hyperparameter tuning, cross-validation, and rigorous testing protocols, e.g., using grid search or random search methods, and validating models with K-fold cross-validation.
- Leveraged predictive modeling to forecast sales trends, predict stock price movements, and detect anomalies, e.g., using ARIMA models for time-series forecasting, LSTM networks for stock price prediction, and anomaly detection algorithms in network security.
- Engineered personalized recommendation systems based on content and collaborative filtering methods, e.g., building recommendation engines for e-commerce platforms or streaming services.
- Implemented accurate image classification and object detection algorithms, e.g., using TensorFlow or PyTorch for classifying images or detecting objects in autonomous driving systems.
- Employed data-driven approaches to identify cybersecurity threats and detect fraudulent activities, e.g., applying machine learning algorithms for intrusion detection systems or transaction monitoring to flag fraudulent transactions and genetic algorithms for optimizing balance and matchmaking in games.
- Implemented an Artificial Intelligence system - Retrieval Augmented Generation (RAG), e.g., Conversational AI (Llama 3) to answer questions about the content of an uploaded PDF, Streamlit-based chatbot application that uses the Llama3 model from the Ollama library to generate and display responses to user-input prompts, and Streamlit-based data analysis application that uses the local Meta Llama 3 model to enable interactive querying and analysis of user-uploaded CSV or XLSX files using Pandas AI and SmartDataframe.

Outlier AI - San Francisco, USA**Software Engineering for AI Training Data (Python) - Independent Contractor (Remote)***Jun 2024 – Nov 2024***Role and Key achievements:**

- Using machine learning and data science knowledge to help and make AI training processes better.
- Describing complicated science and coding ideas unambiguously and in a way that is also enjoyable so that the performance and training of the model are improved.
- Determining the validity and effectiveness of the code, and its compliance with high academic standards of the readability of enterprise software are the tasks of the AI quality assurance specialist.
- Proved to be competent in writing Python code that is efficient and can be maintained in a wide range of applications, such as algorithm and data structure optimization, data science analytics, and AI/ML model development.
- Handle different problems related to coding, including the identification of the best way of implementing Python code and the improvement of algorithms and data structures, data science and data analytics, AI/ML and improving chatbot responses.
- Creating software test cases and debugging that are strong enough to test the efficiency and effectiveness of the generated code, and the code also remains reliable and achieves the expected performance.
- Compose clear instructions with the details necessary to carry out the training and improvement of AI models.
- Use principles of RLHF (Reinforcement Learning from Human Feedback) to offer constructive human feedback and assist in AI code generation through iterative learning processes.
- I recast the scripts of Python where the algorithm of AI/ML/DL was made more efficient up-to 15% at the same time the model's accuracy was unaffected by reducing the computational waste.

Direktorat Jenderal Pajak RI/Indonesian Directorate General of Taxes - Central Java, Indonesia**Data Analyst - Internship (Onsite)***Nov 2019 – Dec 2019***Role and Key achievements:**

- Done the data entry and had the verification for many different financial revenue that was sure about the accuracy and reliability of the records.
- From the financial revenue data, the trends and patterns were found by the data analysis, and it helped with the process of making decisions based on the new information.
- Prepared extensive annual financial reports on the income, thus dealing with strategic planning and organizational transparency was no longer an issue.

Partai Amanat Nasional - Central Java, Indonesia**Data Warehouse Administrator - Freelance (Onsite)***May 2017 – Oct 2017***Role and Key achievements:**

- Conducted very detailed data verification and also accepted the verification process of Party's member population data by KPUD in compliance with their regulations, guaranteeing the safety and security of the data.
- Designed and looked over a database that was organized, thus the data management and organizational needs were simplified and data were on the hand accessibly.
- Contributed to a great extent to the database of the Party by changing and correcting the members who were wrongly categorized henceforth this led to the high accuracy and reliability of the party's database for the electoral process.

Role and Key achievements:

- Participated and assisted in the survey on food commodities which collected accurate and precise data on themes such as the proper materials, corn, and the like for BPS.
- Facilitated the process of the analysis of food commodity data, which was instrumental in detecting the trends and the insights supporting decision-making, and forming good policies were vital.
- Participated in the spatial survey, more precisely, in the areas of maize cultivation, where they lent a hand in generating the geographic data which is the backbone of the planning and the development of the agricultural sector.

EDUCATION**Institut Teknologi Sepuluh Nopember - Surabaya, Indonesia****Master's Degree of Statistics**

Jan 2022 – Apr 2024

- Demonstrated proficiency in conducting research by devising an innovative approach for scaling a previously thought non-scalable community detection algorithm, [published](#) in a peer-reviewed journal.
- Excelled in academic project, demonstrating skills in algorithm design, data structures, and machine learning.
- Completed highly demanding coursework in Statistical Process Control, Econometrics, Inference Statistics, Meta Analysis, Data Analysis, and Intensive Computational Statistics.

Universitas Muhammadiyah - Semarang, Indonesia**Bachelor's Degree of Statistics**

Sep 2019 – May 2021

- Actively participated in statistical seminars, workshops, and conferences, enhancing both theoretical knowledge and practical skills in the field, with a focus on Python and R.
- Excelled in academic projects, demonstrating skills in time series analysis and exploratory data analysis, including techniques requiring graphical visualization for insightful data interpretation and presentation.
- Completed highly demanding coursework in Regression Analysis, Multivariate Analysis, Time Series Analysis, Categorical Data Analysis, and Financial Statistics.

Akademi Statistika Muhammadiyah Semarang - Semarang, Indonesia**Diploma III or Associate's Degree of Statistics**

Sep 2016 – Sep 2019

- Actively participated in student organizations and volunteering initiatives, demonstrating leadership skills and a commitment to community engagement through statistical applications.
- Excelled in academic project, demonstrating skills in data mining, data structures, and their implementation using SQL, including proficient querying, database design, and optimization techniques for efficient data retrieval and manipulation.
- Completed highly demanding coursework in Probability Theory, Stochastic Processes, Bayesian Methods, Experimental Design, Calculus, Linear Algebra, Inventory Statistics, and Data Mining.

ORGANIZATION EXPERIENCE**Ikatan Himpunan Mahasiswa Statistika Indonesia/Association of Indonesian Statistics Students****Member, Zone 3 Central Java and Yogyakarta**

Jan 2018 – Dec 2018

- Get more directly involved in a wide range of events, meetings, and initiatives organized by the association to not only help it achieve its aims but also to establish relations with other members of the association, e.g., listening to speakers at workshops or seminars and joining in at social events.
- Initiate and create bonds with the other members so that there is an environment to share ideas, experiences and take part in academic and statistical research projects.
- Being a representative of the member's needs and problems in the public sphere, and being a person that truly reflects the values and objectives of the association in the statistical community.

VOLUNTEERING**4th Indonesia Statistics Conference and Olympiad (ISCO)****Event Leader of Indonesia Olympiad of Statistics/IOS (Sub-event ISCO)**

May 2018 – Sep 2018

- Be in charge of scheduling and carrying out the Indonesia Olympiad of Statistics (sub-event ISCO), which includes coordinating the events, assignments of resources, and all the office work necessary to secure the event running smoothly.
- Form the team of the volunteer or committee members that are working on organizing the sub-event, which includes giving tasks, advice, and contributing towards teamwork, and thus reaching the objectives of the event in the best possible way.
- Act as the main communication link between the sub-event team, the participants, the sponsors, and the rest of the event stakeholders, while delivering updates, handling the inquiries or complaints, and creating good relationships are some of the ways to make the experience and the success of the event greater.

RESEARCH AND PUBLICATIONSGoogle Scholar: [Amanatullah Pandu Zenklinov](#)

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| • Stunting Risk Factor Analysis Using Seemingly Unrelated Regression (SUR) Integrated with Machine Learning | 2025 |
| • The Fuzzy-Possibilistic Product Partition c-Means (FPPPCM) algorithm for Clustering the Welfare Levels of Regencies in East Java | 2025 |
| • Utilization of Flora in the School Environment as a Drawing Medium for Reading Children's Creativity | 2025 |
| • Intrusion Detection Systems (IDSs) using Multivariate Control Chart Hotelling's T2 with Dimensional Reduction of Factorial Analysis of Mixed Data (FAMD) and Autoencoder | 2024 |
| • Spatial Temporal Density Based Spatial Cluster of Application with Noise untuk Mengelompokkan Curah Hujan di Jawa Timur | 2024 |
| • Pengelompokkan Kemiskinan Kabupaten/Kota Di Jawa Tengah Menggunakan Algoritma Fuzzy Possibilistic C-Means dan Possibilistic Fuzzy C-Means | 2021 |

COURSE AND CERTIFICATIONS

• Generative AI Engineering with LLMs Specialization	IBM	Apr 2025
• English Level C2 Advanced according to the Common European Framework of Reference (CEFR)	EF SET	Apr 2025
• Webinar Akuntansi dan Matematika 2024	UGM	Dec 2024
• Data-Driven Decision Making: Promoting the Use of Statistics in Making Informed Decisions Across Different Sectors Such as Economy, Education, Healthcare, and Governance	IHMSI	Sep 2024
• English Level C1 Advanced according to the Common European Framework of Reference (CEFR)	EF SET	Jul 2024
• Introduction to Machine Learning on AWS	AWS	Jul 2024
• GenAI Chatbots: Create and Deploy OpenAI-Powered Chatbots	Coursera	Jun 2024
• Generative AI with Vertex AI: Build a customer chatbot	Coursera	Jun 2024
• Google Cloud Big Data and Machine Learning Fundamentals	Google	May 2024
• Working with Big Data	Coursera	May 2024
• IBM Machine Learning Professional Certificate	IBM	Apr 2024
• Data Analysis with Python	IBM	Mar 2024
• Deep Learning with Python	IBM	Mar 2024
• Machine Learning with Python	IBM	Mar 2024
• Prompt Engineering	IBM	Mar 2024
• Python for Data Science	IBM	Mar 2024
• International Conference Presenter of 13 th ISNPINSA	UNDIP	Nov 2023
• Researcher and Research Student Enrichment Program	ITS	Nov 2023